Technical Note

Power Electronics and Control Systems

# L(RC) Filter Design Guidelines

## Grid-side filters for active front ends and output filters for induction and permanent-magnet motor drives

*Topologies, sizing relations, resonance placement, damping and interaction with the current control*


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**Abstract:** Design guidelines for the passive filters of two-level voltage-source converters: the L, LC, LCL and L(RC)L filter between an active front end (AFE) and a grid of given impedance, and the LC or L(RC) output filter between an inverter and an induction motor (IM) or a permanent-magnet synchronous motor (PMSM). The note first settles the switching ripple of the converter-side inductor, with exact closed forms for sinusoidal, space-vector and discontinuous PWM, the mean and the rms ripple over the fundamental period, and a comparison with the usual $V_{dc}/(8 f_{sw} L)$ rule. It then gives, for the grid side, the sizing of the converter inductor, of the capacitor (reactive power), of the grid inductor (attenuation at the switching frequency against IEEE 519), the placement of the resonance with respect to the grid frequency, the switching frequency and the critical frequency $f_s/6$ of the digital current loop, the passive damping options (series resistor, R–C branch) with their losses, and the interaction with the grid impedance. For the motor side it covers the reflected-wave overvoltage with long cables, the $dv/dt$ and sine filter options, the resonance window, the loaded resonance set by the high-frequency inductance of the machine (transient inductance of the IM, $L_d/L_q$ of the PMSM), the damping, the capacitor current and voltage budget, and the consequences for the current control and for sensorless (observer-based) control. Two sizing paths with worked examples (400 V/250 kVA AFE; 110 kW IM and PMSM drives) and a bibliography close the note; the numbers are reproduced by the companion Python script.


Revision history

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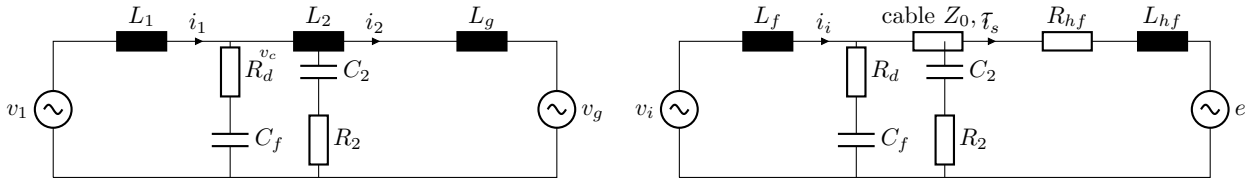
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# 1 Scope and notation

Two-level three-phase voltage-source converter (VSC) [16, 15] with DC link $V_{dc}$, switching frequency $f_{sw}$ ($\omega_{sw} = 2\pi f_{sw}$), double-edge regular-sampled PWM with sampling frequency $f_s = f_{sw}$ (single update) or $2f_{sw}$ (double update). Two applications are treated with the same formulas: the grid side of an active front end (AFE), Fig. 1(a), and the output side of a motor inverter, Fig. 1(b). All quantities are per phase of the star equivalent; $V$ is the RMS line-to-line voltage, $V_{ph} = V/\sqrt{3}$, $I_n$ the rated RMS phase current, $\hat{I}_n = \sqrt{2}I_n$. The base impedance is $Z_b = V^2/S_n$, $L_b = Z_b/\omega_1$, $C_b = 1/(\omega_1 Z_b)$ with $\omega_1 = 2\pi f_1$ the fundamental (grid or motor) angular frequency. Formulas are quoted without derivation except where the derivation is short and not found in the references.




(a) grid side: L ($C_f = 0$), LCL, L(RC)L (b) motor side: LC, L(RC); $L_{hf} = \sigma L_s$ (IM), $L_d, L_q$ (PMSM)

Figure 1: Filter structures. The damping is either the series resistor $R_d$ (dashed) or the parallel branch $C_2$–$R_2$ (the "RC" of L(RC)); only one of the two is used.

# 2 Switching ripple of the converter-side inductor

The converter-side inductor ($L_1$ or $L_f$) is sized first from its switching ripple. With the capacitor node a stiff voltage at $f_{sw}$ (LCL, LC) or a stiff grid (plain L), the phase current ripple over one switching period $T_{sw} = 1/f_{sw}$ is piecewise linear with slopes $(v_{1,n} - v_{ref})/L$, where $v_{1,n}$ is the converter phase voltage with respect to the load neutral (0, $\pm V_{dc}/3$, $\pm 2V_{dc}/3$) and $v_{ref}$ the reference. Let $m = \hat{V}_{ref}/(V_{dc}/2)$ be the modulation index ($m \le 1$ for sinusoidal PWM, $m \le 2/\sqrt{3} = 1.155$ for space-vector or third-harmonic PWM). All ripple figures scale as

$$\Delta i = k\,\frac{V_{dc}}{f_{sw}L}, \tag{2.1}$$

and the coefficient $k$ depends on the modulator, on $m$ and on the position $\theta$ along the fundamental period. Two closed forms cover the two extremes (phase voltage at its peak, $\theta = 0$, and at its zero crossing, $\theta = 90°$):

$$k_{\mathrm{pp}}(\theta = 0) = \frac{m}{4}\Big(1 - \frac{m}{2}\Big)\ \text{(SPWM)}, \qquad \frac{m}{4}\Big(1 - \frac{3m}{4}\Big)\ \text{(SVM, THI)}, \qquad \frac{m}{2}\Big(1 - \frac{3m}{4}\Big)\ \text{(DPWM, clamped phase)}, \tag{2.2}$$

$$k_{\mathrm{pp}}(\theta = 90°) = \frac{\sqrt{3}\,m}{12} = \frac{m}{6.93} \qquad \text{(all modulators)}. \tag{2.3}$$

Eq. (2.2) follows from $\Delta i_{\mathrm{pp}} = v_{ref}\,(T_0 + |T_{0a} - T_{0b}|)/(2L)$, with $T_0 = T_{0a} + T_{0b}$ the total zero-vector time and $T_{0a}$, $T_{0b}$ its split between 000 and 111 (equal for SVM, $(1-m)T_{sw}/2$ and $(1-m/2)T_{sw}/2$ for SPWM at $\theta = 0$, all on one vector for the clamped phase of DPWM); at $\theta = 0$, $T_0/T_{sw} = 1 - 3m/4$. Eq. (2.3) follows from the two active vectors of equal dwell $T_1 = T_2 = (\sqrt{3}/4)\,m\,T_{sw}$ giving $\pm V_{dc}/3$ to the phase at its zero crossing. The peak-to-peak ripple is largest at the phase peak for small $m$ and at the zero crossing for large $m$ (crossover at $m = 0.56$ for SVM, 0.85 for SPWM). The worst case over the whole linear range is therefore

$$\Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{4\sqrt{3}\,f_{sw}L} = \frac{V_{dc}}{6.93\,f_{sw}L} \qquad \text{(SPWM, at } m = 1\text{)}, \tag{2.4}$$

$$\Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{6\,f_{sw}L} \qquad \text{(SVM/THI, at } m = 2/\sqrt{3}\text{; DPWM, at } m = 2/3 \text{ and } 2/\sqrt{3}\text{)}.$$

The RMS value of the ripple over the fundamental period, needed for the inductor and capacitor losses and for the harmonic current injected into the capacitor, is given by the harmonic distortion factor of Holmes and Lipo

[15]:

$$I_{rip,rms} = \frac{V_{dc}}{24\, f_{sw} L}\sqrt{\mathrm{HDF}(m)}, \qquad \mathrm{HDF} = \begin{cases} \frac{3}{2}m^2 - \frac{4\sqrt{3}}{\pi}m^3 + \frac{9}{8}m^4 & \text{(SPWM)},\\ \frac{3}{2}m^2 - \frac{4\sqrt{3}}{\pi}m^3 + \left(\frac{27}{16} - \frac{81\sqrt{3}}{64\pi}\right)m^4 & \text{(SVM)}. \end{cases} \tag{2.5}$$

Table 1 lists $k$ for the maximum and the mean (over the fundamental period) peak-to-peak ripple and for the RMS ripple, computed exactly by the companion script; the closed forms (2.2)–(2.5) are reproduced to four digits.

Table 1: Ripple coefficient $k$ of (2.1), written as $1/x$: maximum peak-to-peak over the fundamental period, mean peak-to-peak, RMS of the ripple. Two-level VSC, double-edge regular sampling, isolated neutral.

| | SPWM | | | SVM / THI | | | DPWM1 (60° clamping) | | |
|---|---|---|---|---|---|---|---|---|---|
| $m$ | max pp | mean pp | RMS | max pp | mean pp | RMS | max pp | mean pp | RMS |
| 0.4 | 1/12.5 | 1/17.6 | 1/67 | 1/14.3 | 1/18.7 | 1/68 | 1/7.1 | 1/10.4 | 1/35 |
| 0.6 | 1/9.5 | 1/13.7 | 1/52 | 1/11.6 | 1/15.3 | 1/55 | 1/6.1 | 1/8.8 | 1/30 |
| 0.8 | 1/8.3 | 1/12.1 | 1/44 | 1/8.7 | 1/14.4 | 1/49 | 1/6.3 | 1/8.9 | 1/31 |
| 0.9 | 1/7.7 | 1/11.4 | 1/41 | 1/7.7 | 1/14.0 | 1/47 | 1/6.8 | 1/9.4 | 1/32 |
| 1.0 | 1/6.9 | 1/10.9 | 1/37 | 1/6.9 | 1/13.5 | 1/45 | 1/6.9 | 1/10.4 | 1/35 |
| 1.1 | – | – | – | 1/6.3 | 1/13.0 | 1/42 | 1/6.3 | 1/11.5 | 1/37 |
| 1.155 | – | – | – | 1/6.0 | 1/12.8 | 1/40 | 1/6.0 | 1/11.9 | 1/38 |

**Which ripple formula to use.** The widely used $L = V_{dc}/(8 f_{sw}\Delta i_{\mathrm{pp}})$ is neither the maximum nor the mean: at $m \approx 1$ with SVM the true maximum is $V_{dc}/(6.9 f_{sw}L)$ (15 % above the rule), the worst case over $m$ is $V_{dc}/(6 f_{sw}L)$ (33 % above, the value used by Liserre et al. [1] and Reznik et al. [2]), while the mean peak-to-peak over the fundamental is $V_{dc}/(13.5 f_{sw}L)$ (40 % below). The rule is a fair "typical period" estimate; it is not an upper bound. Recommended use: (i) size $L$ and check the saturation current $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$ with the maximum of Table 1 at the largest $m$ of the application ($V_{dc}/(6 f_{sw}L)$ when the converter reaches the hexagon, $V_{dc}/(7 f_{sw}L)$ for an AFE at $m \approx 1$); (ii) use the RMS column, or (2.5), for the capacitor current, the damping losses and the harmonic current at $f_{sw}$; (iii) with discontinuous PWM at the same $f_{sw}$ the ripple is up to twice that of SVM at low $m$, so compare at equal switching losses ($f_{sw}$ raised by 1.5).

# 3 Filter structures and transfer functions

With $Z_1 = sL_1$, $Z_2 = s(L_2 + L_g)$ and $Z_c$ the impedance of the shunt network of Fig. 1(a), and the grid voltage a short at the switching harmonics,

$$\frac{i_1}{v_1} = \frac{Z_c + Z_2}{Z_1 Z_2 + Z_c(Z_1 + Z_2)}, \qquad \frac{i_2}{v_1} = \frac{Z_c}{Z_1 Z_2 + Z_c(Z_1 + Z_2)}, \qquad \frac{i_2}{i_1} = \frac{Z_c}{Z_c + Z_2}. \tag{3.1}$$

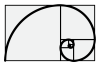
For the undamped LCL ($Z_c = 1/(sC_f)$), writing $L_2' = L_2 + L_g$,

$$\frac{i_2}{v_1} = \frac{1}{s^3 L_1 L_2' C_f + s(L_1 + L_2')}, \qquad \omega_{res}^2 = \frac{L_1 + L_2'}{L_1 L_2' C_f}, \qquad \frac{i_2}{i_1} = \frac{1}{1 + s^2 L_2' C_f}, \qquad \omega_2^2 = \frac{1}{L_2' C_f}. \tag{3.2}$$

At the switching frequency the attenuation of the LCL with respect to its own converter current is

$$k_a = \left|\frac{i_2}{i_1}\right|_{\omega_{sw}} = \frac{1}{\omega_{sw}^2 L_2' C_f - 1} \approx \left(\frac{f_2}{f_{sw}}\right)^2, \qquad \text{equivalently [1]} \quad \frac{i_2(\omega_{sw})}{i_L(\omega_{sw})} = \frac{1}{\left|1 + r\,(1 - \omega_{sw}^2 L_1 C_f)\right|}, \quad r = \frac{L_2}{L_1}, \tag{3.3}$$

the second form being referred to the current $i_L = v_1/(j\omega_{sw}L_1)$ of the plain L filter; the two coincide where $C_f$ is a short at $f_{sw}$. The damped networks are $Z_c = R_d + 1/(sC_f)$ (series resistor) and $Z_c = \frac{1}{sC_1} \parallel \left(R_2 + \frac{1}{sC_2}\right)$ (R–C branch, with $C_1 + C_2$ the total capacitance). The series resistor turns the −60 dB/decade roll-off of $i_2/v_1$ into −40 dB/decade above $1/(2\pi R_d C_f)$ and degrades (3.3) to $k_a = \sqrt{R_d^2 + X_c^2}/\sqrt{R_d^2 + (\omega_{sw}L_2' - X_c)^2}$, $X_c = 1/(\omega_{sw}C_f)$; the R–C branch keeps the roll-off and is analysed in [14, 11, 12, 13]. The motor-side LC filter of Fig. 1(b) obeys the same algebra with $Z_1 = sL_f$ and $Z_2 = R_{hf} + sL_{hf}$ (the back-EMF is a short at the ripple



frequencies, the cable adds its capacitance to $C_f$):

$$\frac{v_c}{v_i} = \frac{Z_c \parallel Z_2}{Z_1 + Z_c \parallel Z_2}, \qquad f_0 = \frac{1}{2\pi\sqrt{L_f C_f}}, \qquad f_p = f_0\sqrt{1 + \frac{L_f}{L_{hf}}}, \qquad \left|\frac{v_c}{v_i}\right|_{f_{sw}} \approx \frac{1}{(f_{sw}/f_p)^2 - 1}, \tag{3.4}$$

where $f_0$ is the unloaded resonance and $f_p$ the resonance with the machine connected.

# 4 Grid-side filter of an active front end

## 4.1 Design inputs

Rated power $S_n$, voltage $V$, frequency $f_1$, DC-link voltage $V_{dc}$ (boost operation requires $V_{dc} \geq \sqrt{2}V(1+\epsilon_L)$, i.e. $m \leq 2\sqrt{2}V_{ph}(1+\epsilon_L)/V_{dc}$ with $\epsilon_L$ the filter drop), $f_{sw}$ and $f_s$, the current-feedback choice (converter current $i_1$ or grid current $i_2$), the range of the grid impedance at the point of common coupling (PCC), and the harmonic limits. The grid inductance per phase, transformer included, is

$$L_g = \frac{V^2}{\omega_1 S_{sc}} = \frac{u_k V^2}{\omega_1 S_T} + L_g^{MV}, \qquad \frac{S_{sc}}{S_n} = \frac{I_{sc}}{I_L} \text{ (short-circuit ratio)}, \tag{4.1}$$

to be bracketed between the stiffest ($S_{sc,max}$) and the weakest ($S_{sc,min}$) expected PCC. The limits of IEEE 519 [17] at the PCC depend on $I_{sc}/I_L$: for $I_{sc}/I_L < 20$ the individual current harmonics are limited to 4 % ($h < 11$), 2 % ($11 \leq h < 17$), 1.5 % ($17 \leq h < 23$), 0.6 % ($23 \leq h < 35$) and 0.3 % ($35 \leq h \leq 50$) of $I_L$, with TDD 5 %; the switching-frequency cluster of an AFE lies above $h = 50$, where the 0.3 % value is used as design practice. IEC 61800-3 [18] sets the EMC environment (C1–C4) and refers to the IEC 61000-3 series for the low-frequency emission.

## 4.2 Converter-side inductor $L_1$

From Section 2, with the ripple allowance $k_r = \Delta i_{\mathrm{pp},max}/\hat{I}_n$ (15–30 % is usual; a smaller value increases $L_1$, its losses and the voltage drop, a larger one the switching losses and the capacitor current),

$$L_1 \geq \frac{k\,V_{dc}}{f_{sw}\,k_r \hat{I}_n}, \qquad k = \frac{1}{7}\ (\text{SVM},\ m \approx 1), \quad \frac{1}{6}\ (\text{worst case}), \tag{4.2}$$

and the drop $\epsilon_L = \omega_1(L_1 + L_2)I_n/V_{ph}$ (typically 0.05–0.15 pu in total) must be compatible with $V_{dc}$. The inductor carries $I_n$ plus the ripple, must not saturate at $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$ (plus the overload and the inrush at pre-charge), and its core loss is set by the ripple RMS of Table 1.

## 4.3 Capacitor $C_f$

The capacitor is bounded by its reactive power at the fundamental, which the converter must compensate (and which raises the no-load current),

$$C_f \leq x\,C_b = \frac{x\,S_n}{\omega_1 V^2}, \qquad Q_c = \omega_1 C_f V^2 = xS_n, \qquad x \leq 0.05\ [1], \tag{4.3}$$

($C_f$ is the star value; a delta bank uses $C_f/3$ per branch). A larger $C_f$ lowers $L_2$ for a given attenuation and lowers $f_{res}$; with $x$ = 2–5 % the capacitor is the cheapest element and is normally taken at its upper bound. The capacitor must withstand $1.1V$ plus the switching ripple voltage $\hat{v}_{c,sw} \approx I_{rip,rms}\sqrt{2}/(\omega_{sw}C_f)$, carry $I_c = \sqrt{(\omega_1 C_f V_{ph})^2 + I_{rip,rms}^2}$ and be rated for the harmonic background of the grid (5th, 7th) that it absorbs through $L_2'$.

## 4.4 Grid-side inductor $L_2$ and attenuation

The converter-side current at the switching frequency is estimated from the ripple RMS of Table 1 (assigning the whole ripple to the first carrier group is conservative by about 15 %; the sideband spectrum is in [15]); with



the admissible grid harmonic $I_{2,sw} \le 0.003 I_n$ the required attenuation is $k_a = I_{2,sw}/I_{1,sw}$ and, from (3.3) with $L_g = 0$ (worst case: $L_g$ only adds attenuation),

$$L_2 \ge \frac{1 + 1/k_a}{\omega_{sw}^2 C_f}, \qquad r = \frac{L_2}{L_1} = 0.2\text{–}1. \tag{4.4}$$

Values of $r$ above unity give little further attenuation for the volume [1, 3]; the grid inductor may be the leakage of the dedicated transformer or a line reactor. If the attenuation cannot be met, raise $C_f$ up to its bound, then $L_1$ (which lowers $I_{1,sw}$ linearly), then $f_{sw}$.

## 4.5 Resonance placement

With $L_g$ between $L_g^{min}$ and $L_g^{max}$ the resonance (3.2) sweeps the band $[f_{res}^{min}, f_{res}^{max}]$, $f_{res}^{min}$ at the weakest grid. The constraints are

$$10 f_1 \le f_{res}^{min}, \qquad f_{res}^{max} \le \frac{f_{sw}}{2}\ [1], \qquad f_{res} \notin \Big[0.8, 1.2\Big]\frac{f_s}{6}\ [6]. \tag{4.5}$$

The lower bound keeps the resonance away from the low-order grid harmonics and from the current-loop crossover $f_c$ (design $f_c \le f_{res}/3$, with $f_c \approx f_{sw}/20 \ldots f_{sw}/10$ for a PI in the $dq$ frame [7]); the upper bound preserves the attenuation at $f_{sw}$. The third constraint comes from the discrete-time analysis of the current loop with the computation and PWM delay ($1.5T_s$): with converter-current feedback the LCL resonance is stable without any damping only for $f_{res} > f_s/6$, with grid-current feedback only for $f_{res} < f_s/6$, and at $f_{res} = f_s/6$ no controller gain stabilises the loop [6]. Place the whole band $[f_{res}^{min}, f_{res}^{max}]$ on the side consistent with the feedback variable, or provide damping (passive, Section 4.6, or active) whenever the band crosses or approaches $f_s/6$ or when the grid impedance is uncertain [8, 5].

## 4.6 Passive damping

**Series resistor.** The classical choice [1] is one third of the capacitor reactance at the resonance,

$$R_d = \frac{1}{3\omega_{res} C_f}, \qquad \zeta \approx \frac{R_d}{2}\sqrt{\frac{C_f}{L_p}} \approx \frac{1}{6}, \qquad L_p = \frac{L_1 L_2'}{L_1 + L_2'}, \qquad P_{R_d} = 3R_d\big[(\omega_1 C_f V_{ph})^2 + I_{rip,rms}^2\big], \tag{4.6}$$

where the loss has a fundamental term (the whole capacitor current flows in $R_d$) and a switching term; both are usually 0.1–0.3 % of $S_n$ [4]. The price is the attenuation penalty of Section 3.

**R–C branch.** Splitting the capacitance into $C_1$ and $C_2 = nC_1$ with $R_2$ in series with $C_2$ keeps the $-60$ dB/decade slope. With $R_0 = \sqrt{L_p/C_1}$ and $r = R_2/R_0$, the three closed-form choices of [14] are

$$\begin{aligned} r_Z &= \sqrt{\frac{(2+n)(4+3n)}{2n^2(4+n)}} && \text{minimum peak of the output impedance [11],} \\ r_H &= \frac{\sqrt{(1+n)(2+n)/2}}{n} && \text{minimum peak of } |v_c/v_1|, \\ r_P &= \frac{2+n}{2n} && \text{real pole at } \omega_0 = 1/\sqrt{L_p C_1}, \end{aligned} \tag{4.7}$$

with $\zeta = n/(2\sqrt{2(2+n)})$ for $r_P$ (0.20 for $n = 1$, 0.35 for $n = 2$) and a peak of $|v_c/v_1|$ not lower than $(2+n)/n$. On the AC grid the branch carries the fundamental capacitive current of $C_2$, so

$$P_{R_2} = 3R_2\frac{(\omega_1 C_2 V_{ph})^2}{1 + (\omega_1 C_2 R_2)^2} + 3\frac{v_{c,sw}^2}{R_2}, \qquad v_{c,sw} \approx \frac{I_{rip,rms}}{\omega_{sw} C_1}, \tag{4.8}$$

which limits $n$ to 1–2 and makes the loss comparable with the series resistor [4]. Both networks are to be checked at $f_{res}^{min}$ and $f_{res}^{max}$; in modern AFEs passive damping is kept small as a robustness backstop and the bulk of the damping is active (capacitor-current or capacitor-voltage feedback [7, 5, 10]).



## 4.7 Interaction with the grid impedance and stability

The grid sees the converter as a current source with an output admittance $Y_c(s)$ shaped by the current controller; stability of the interconnection is that of the loop $Z_g(s)Y_c(s)$ [9]. With the digital delay the real part of $Y_c$ becomes negative between about $f_s/6$ and $f_s/2$ for converter-current control, so any resonance of $Z_g$ with the filter falling in that band is destabilised, whatever the gains; passive damping and the choice of $f_{res}$ of Section 4.5 move the resonance out of the band or make $Y_c$ passive there. Practical rules: (i) evaluate $f_{res}$ and the phase margin at both ends of the $L_g$ range; (ii) a weak grid (SCR $< 3$) also demands a slow PLL ($\leq 20\,\mathrm{Hz}$) and a reduced current-loop bandwidth; (iii) the background distortion of the grid is amplified by the LCL between $f_2$ and $f_{res}$, which is why $f_{res} \geq 10 f_1$ and why the capacitor must be rated for the 5th/7th harmonics; (iv) verify the design with the transformer leakage and the cable of the actual installation, not with $L_g = 0$ only.

## 4.8 Sizing path and worked example

| Step | Relation | Example: 400 V, 250 kVA, 50 Hz, $V_{dc} = 700\,\mathrm{V}$, $f_{sw} = 4\,\mathrm{kHz}$, $f_s = 8\,\mathrm{kHz}$, SVM, $i_1$ feedback |
|---|---|---|
| G1 | base values, $L_g$ range from (4.1) | $I_n = 361\,\mathrm{A}$, $Z_b = 0.64\,\Omega$, $L_b = 2.04\,\mathrm{mH}$, $C_b = 4.97\,\mathrm{mF}$; $S_{sc} =$ 5–20 MVA ($I_{sc}/I_L = 20 \ldots 80$) $\Rightarrow L_g = 25$–$102\,\mu\mathrm{H}$ |
| G2 | $L_1$ from (4.2), $k_r = 0.25$ at the rated $m$ | $m = 0.98$: $k = 1/7.07$, $L_1 \geq 194\,\mu\mathrm{H} \Rightarrow L_1 = 200\,\mu\mathrm{H}$ (9.8 %): $\Delta i_{\mathrm{pp},max} = 124\,\mathrm{A}$ (24 %), mean 64 A, $I_{rip,rms} = 19.2\,\mathrm{A}$; worst case $V_{dc}/(6 f_{sw} L_1) = 146\,\mathrm{A}$, rule $V_{dc}/(8 f_{sw} L_1) = 109\,\mathrm{A}$ |
| G3 | $C_f$ from (4.3), $x = 0.05$ | $C_f \leq 249\,\mu\mathrm{F} \Rightarrow C_f = 250\,\mu\mathrm{F}$ star (83 µF delta), $Q_c = 12.6\,\mathrm{kvar}$ |
| G4 | $k_a$ and $L_2$ from (4.4) | $I_{2,sw} \leq 0.003 I_n = 1.08\,\mathrm{A}$, $I_{1,sw} \approx 19.2\,\mathrm{A} \Rightarrow k_a \leq 0.056$ ($-25\,\mathrm{dB}$), $L_2 \geq 119\,\mu\mathrm{H} \Rightarrow L_2 = 125\,\mu\mathrm{H}$ ($r = 0.62$), $k_a = 0.053$, $I_{2,sw} = 1.03\,\mathrm{A}$ (0.28 %); $L_1 + L_2 = 16\,\%$ |
| G5 | $f_{res}$ from (3.2), band (4.5) | $f_{res} = 1148\,\mathrm{Hz}$ ($L_g = 0$), 976–1086 Hz over the $L_g$ range; $10 f_1 =$ 500 Hz, $f_{sw}/2 = 2\,\mathrm{kHz}$; $f_s/6 = 1333\,\mathrm{Hz}$: the band is below $f_s/6$ with $i_1$ feedback $\Rightarrow$ damping required (or $i_2$ feedback) |
| G6 | damping (4.6) or (4.7)–(4.8) | series: $R_d = 185\,\mathrm{m\Omega}$, $\zeta \approx 0.17$, $P = 387\,\mathrm{W}$ (0.15 %), $k_a$ degraded to 0.082; R–C with $n = 1$: $C_1 = C_2 = 125\,\mu\mathrm{F}$, $L_p = 77\,\mu\mathrm{H}$, $R_0 =$ $0.78\,\Omega$, $r_P = 1.5 \Rightarrow R_2 = 1.18\,\Omega$, $\zeta = 0.20$, $P = 290 + 95 = 385\,\mathrm{W}$ (0.15 %), slope preserved |
| G7 | ratings | $L_1$: 361 A, saturation $\geq 583\,\mathrm{A}$, ripple 19 A RMS at 4 kHz; $L_2$: line reactor 361 A; $C_f$: $\geq 440\,\mathrm{V}$ AC, $I_c \approx 26\,\mathrm{A}$ |
| G8 | control checks | $f_c \leq f_{res}^{min}/3 \approx 320\,\mathrm{Hz}$ ($\approx f_{sw}/12$); verify phase margin at $L_g^{min}$ and $L_g^{max}$ with the delay $1.5 T_s$; PLL $\leq 20\,\mathrm{Hz}$ at $S_{sc,min}$ |

The example shows the typical outcome at 4 kHz: a 0.3 % target at the switching cluster needs $L_1 + L_2 \approx 0.15\,\mathrm{pu}$ with the capacitor at its 5 % bound, and the resonance lands at $f_{res}/f_{sw} \approx 0.25$–$0.3$, i.e. below the critical $f_s/6$ with double update. Raising $f_{sw}$ to 8–10 kHz halves the inductances and moves $f_{res}$ above $f_s/6$ at the same $x$; with grid-current feedback the present placement is instead the inherently damped one.

# 5 Output filters between inverter and motor

## 5.1 Why: $dv/dt$, reflected waves and insulation stress

A PWM edge with rise time $t_r$ travels along the motor cable (surge impedance $Z_0 = \sqrt{L'/C'} \approx 20$–$80\,\Omega$, velocity $v \approx 150$–$200\,\mathrm{m/\mu s}$, transit time $\tau = l/v$) and is reflected at the motor with coefficient $\Gamma_m = (Z_m - Z_0)/(Z_m + Z_0)$, close to +1 because the high-frequency impedance of the winding is $\gg Z_0$ (0.8–0.95 for motors below 100 kW), and at the inverter with $\Gamma_s \approx -1$. The terminal voltage reaches $V_{dc}(1 + \Gamma_m)$ when the edge is shorter than the round trip [23, 24]:

$$l_{crit} = \frac{v\, t_r}{2}, \qquad \hat{V}_m \leq V_{dc}\Big[1 + \Gamma_m \min\Big(1, \frac{l}{l_{crit}}\Big)\Big], \tag{5.1}$$

the second expression being the design envelope of [25] (the lossless-line solution for a linear edge lies below it; real edges and cable losses smooth it). Double pulsing and polarity reversal within the cable ringing period $4\tau$ raise the peak up to $3V_{dc}$ [26]; the high-frequency models needed for prediction are in [27]. With IGBT edges of 0.1–0.3 µs, $l_{crit}$ is 10–30 m and any industrial cable gives the full $2V_{dc}$. The limits are those of the motor insulation: NEMA MG 1 Part 31 [22] (definite-purpose inverter-fed motors) allows $3.1\,V_{rated}$ peak with $t_r \geq 0.1\,\mu\mathrm{s}$ for $\leq 600\,\mathrm{V}$ motors, Part 30 (general purpose) 1000 V peak and $t_r \geq 2\,\mu\mathrm{s}$; IEC 60034-18-41 [20]



qualifies the insulation by stress category (overshoot factor and impulse rise time), IEC TS 60034-25 [21] gives the application guide and IEC TS 61800-8 [19] the method to specify the voltage at the power interface. The common-mode $dv/dt$ (bearing currents, EMI) is a separate issue, mitigated by common-mode chokes and not covered here.

## 5.2 Filter options

- **Output reactor** (1–3 %): slows the edge seen by the cable only through the cable capacitance, $t_r \approx 2.2\sqrt{LC_{cable}}$-like; mostly limits the capacitive charging current and the inverter stress; insufficient alone above 50–100 m.
- $dv/dt$ **filter** (LC, $f_0 = 5\text{–}10\, f_{sw}$, damped): limits the edge at the filter output to $t_r \approx 0.3/f_0$ (undamped 10–90 % rise; 0.3–0.5 with damping). From (5.1), for an admissible overshoot $OF = \hat{V}_m/V_{dc} - 1$ and cable length $l$: $l_{crit} \geq l\,\Gamma_m/OF$, i.e. $t_r \geq 2l\Gamma_m/(v\,OF)$. The capacitor is charged and discharged twice per switching period, so irrespective of the damping resistor the loss is bounded by $P \approx 3 f_{sw} C_f V_{dc}^2$: $C_f$ must stay at 0.1–0.5 µF and the filter cannot be made slow. Damping $\zeta = 0.3\text{–}0.5$ with $R_d$ in series with $C_f$ or a resistor across $L_f$ (carries no fundamental); clamping diodes to the DC bus recover the energy [28, 25].
- **Sine filter** (LC or L(RC), $f_0 \ll f_{sw}$): sinusoidal motor voltage, no reflection problem at any length (the edge is $\sim 1/(2f_0)$), lower motor noise and harmonic losses, standard motors usable, but fundamental drop, capacitor current, reduced control bandwidth and field-weakening range [29, 28, 32]. The remainder of this section is on the sine filter.
- **Motor-terminal RC** ($R = Z_0$, $C \approx 3t_r/Z_0$): matches the cable at the motor end [24]; used when the filter cannot be at the inverter.

## 5.3 Sine filter: resonance window

The unloaded resonance $f_0$ must sit between the highest fundamental and its low-order harmonics on one side, and the switching frequency on the other:

$$(8\ldots10)\, f_{1,max} \leq f_0 \leq \frac{f_{sw}}{4\ldots3}, \qquad \left|\frac{v_c}{v_i}\right|_{f_{sw}} \approx \left(\frac{f_p}{f_{sw}}\right)^2 = -20\text{–}-28\,\text{dB}. \tag{5.2}$$

The lower bound keeps the 5th and 7th harmonics generated by the dead time and by the inverter nonlinearity (at $5f_1$, $7f_1$, up to 1.4 kHz for a 200 Hz machine) away from the resonance, where they would be amplified by $Q = 1/(2\zeta)$, and keeps the current-loop crossover below $f_0/3$. The window exists only if $f_{sw} \geq (24\ldots40)\, f_{1,max}$: a 50 Hz induction motor in field weakening up to 75 Hz is served by $f_{sw} = 4$ kHz, a high-speed or high-pole PMSM at 200 Hz needs $f_{sw} \geq 6\text{–}8$ kHz or a relaxed attenuation. This, with the capacitor current, is the main reason why sine filters are heavier for PMSM drives.

## 5.4 Sine filter: inductor, capacitor, loaded resonance

**Inductor.** Sized by the fundamental drop at $f_{1,max}$ and checked for the inverter ripple,

$$L_f \leq \frac{\epsilon_L V_{ph}}{\omega_{1,max} I_n}, \quad \epsilon_L = 0.05\text{–}0.10; \qquad \Delta i_{\text{pp},max} = \frac{V_{dc}}{6 f_{sw} L_f} \quad \text{(the inverter reaches } m = 2/\sqrt{3} \text{ at } f_{1,max}\text{)}, \tag{5.3}$$

With $L_f = 3\text{–}6$ % the inverter current ripple is 25–45 % of $\hat{I}_n$: it flows entirely into $C_f$, the motor sees none of it, but the inverter must be rated for the peak and for the switching losses, and the inductor core for the ripple RMS of Table 1. The drop reduces the voltage available to the machine by $\omega_1 L_f I_n$ (vectorially): it must be budgeted in $V_{dc}$ or accepted as an earlier field-weakening onset.

**Capacitor.** From the chosen $f_0$ and the fundamental capacitive current,

$$C_f = \frac{1}{(2\pi f_0)^2 L_f}, \qquad I_c = \omega_{1,max} C_f V_{ph} \leq (0.1\ldots0.2)\, I_n, \qquad \boldsymbol{i}_i = \boldsymbol{i}_s + j\omega_1 C_f \boldsymbol{v}_c, \quad \boldsymbol{v}_i = \boldsymbol{v}_c + (R_{Lf} + j\omega_1 L_f)\, \boldsymbol{i}_i, \tag{5.4}$$

the last two relations giving the inverter current and voltage that set the inverter rating and shift the torque-maximising trajectories of the machine [32]. The capacitor is rated for $1.1V$ plus the switching ripple $\hat{v}_{c,sw} \approx$



$\sqrt{2}I_{rip,rms}/(\omega_{sw}C_f)$ and for $\sqrt{I_c^2 + I_{rip,rms}^2}$.

**Loaded resonance.** With the machine connected the resonance moves from $f_0$ to $f_p = f_0\sqrt{1 + L_f/L_{hf}}$ of (3.4), where $L_{hf}$ is the inductance the machine presents at kilohertz frequencies: the transient (total leakage) inductance $\sigma L_s = L_{ls} + L'_{lr}$ of the induction motor, 0.15–0.25 pu, or $L_d$, $L_q$ of the PMSM, 0.1–0.3 pu for surface magnets and 0.3–0.8 pu for interior magnets with $L_q > L_d$ and a strong dependence on $i_q$ (saturation). The shift is 4–20 % and must be included in (5.2); for an IPM it differs between the $d$ and $q$ axes, so the plant seen by the current controller is anisotropic. The damping contributed by the machine is small because $L_{hf} \gg L_f$ decouples it from the capacitor node: the machine branch acts at $f_p$ as a conductance $G_m = R_{hf}/(R_{hf}^2 + (2\pi f_p L_{hf})^2)$ and gives $\zeta_m = \frac{1}{2}G_m\sqrt{L_p/C_f}$ with $L_p = L_f \parallel L_{hf}$, i.e. a few $10^{-3}$ even with the deep-bar rotor resistance of the IM (0.05–0.15 pu at 1 kHz), and $10^{-4}$ for the PMSM, whose rotor has no cage. The resonance therefore has $Q \approx 10$–$20$, set by the inductor losses, unless it is damped explicitly.

## 5.5 Sine filter: damping

The same two passive networks of Section 4.6 apply with $\omega_{res} \to 2\pi f_p$ and $L_p = L_f \parallel L_{hf}$; on the motor side the fundamental reaches $f_{1,max}$ and the capacitor is a larger fraction of $C_b$, so the fundamental term dominates the loss of both networks and the R–C branch, which adds $C_2 = nC_f$ to the capacitance, costs typically twice the series resistor. The series resistor $R_d = 1/(3 \cdot 2\pi f_p C_f)$ ($\zeta \approx 0.17$) is the usual passive choice at 0.5–1 % of $P_n$; it also reduces the attenuation by the factor $\sqrt{1 + (2\pi f_{sw} R_d C_f)^2}$. Active damping (a virtual resistor through capacitor-voltage or capacitor-current feedback, or the cascaded control of the next section) is lossless and is preferred where the control has the necessary measurements; the passive resistor is then kept small as a backstop for the V/f mode and for the resonance excited by the dead-time harmonics.

## 5.6 Consequences for the control, IM versus PMSM

With the filter the inverter current $i_i$ and voltage $v_i$ are no longer the stator quantities: $i_s = i_i - C_f\, dv_c/dt$ and $v_s = v_c = v_i - (R_{Lf} + L_f\, d/dt)\, i_i$. A vector controller that ignores the filter commits a steady-state error of 5–15 % on the voltage (the drop on $L_f$) and of 5–20 % on the current (the capacitor current), and has the undamped LC resonance inside its loop. The structures that work are:

- **Cascaded control** [30]: inner loop on $i_i$ (plant $L_f$, bandwidth $\alpha_i \le 2\pi f_{sw}/10$), middle loop on $v_c$ (plant $C_f$, $\alpha_v \approx \alpha_i/3 \ldots \alpha_i/5$), outer loop on $i_s$ (plant $L_{hf}$, $\alpha_s \approx \alpha_v/3 \ldots \alpha_v/5$); the inner loops damp the resonance actively. The stator-current bandwidth ends at $f_p/5$–$f_p/3$, i.e. 100–400 Hz instead of the 0.5–1 kHz of a filterless drive, and the speed loop is correspondingly slower; this is the price for servo-like PMSM applications.
- **Full-order observer** of the filter and machine states from $i_i$ (and $v_i$ reference), which restores $i_s$, $v_s$ and the flux without extra sensors [30, 31]; the observer must include $L_f$, $C_f$ and the correct $L_{hf}$ (for the IPM, $L_d$ and $L_q$ with their saturation).
- **Direct measurement** of $i_s$ or $v_c$ after the filter when the cabinet layout allows it.

**Sensorless control.** Back-EMF (fundamental-model) observers of the PMSM need $v_s$: using $v_i$ instead of $v_c$ injects the $L_f$ drop directly into the position estimate, so the filter model must be in the observer [31]; the same holds for the IM flux observer [30]. Saliency-based high-frequency injection ($f_{inj}$ = 0.5–2 kHz) is attenuated and phase-shifted by the filter and, near $f_p$, the response is dominated by the filter rather than by the saliency: keep $f_{inj} \le f_p/3$ with the filter transfer function compensated, or measure $i_s$ directly; many drives disable injection with a sine filter.

**IM versus PMSM, summary.** (i) $L_{hf}$: $\sigma L_s$ for the IM, $L_d/L_q$ for the PMSM, larger and anisotropic for the IPM, so the loaded resonance shift and the plant differ per axis. (ii) The IM rotor cage adds a lossy branch but, as shown, the damping it gives to the filter is marginal; the PMSM gives none. (iii) The PMSM usually has the higher $f_{1,max}$, which narrows the window (5.2) and raises the required $f_{sw}$. (iv) The filter drop and the capacitor current change the voltage and current limits: for the PMSM they shift the MTPA/MTPV trajectories and anticipate the field weakening [32]; for the IM they reduce the maximum torque in field weakening. (v) The benefit differs too: for the PMSM the sinusoidal current removes the magnet eddy-current losses and the demagnetisation risk from the switching ripple at high temperature; for the IM it removes the rotor-bar harmonic losses and the acoustic noise.

## 5.7 Sizing path and worked example

| Step | Relation | IM 110 kW, 4-pole, 400 V, 195 A, $f_{1,max} = 75$ Hz, $\sigma L_s = 0.18$ pu, $f_{sw} = 4$ kHz | PMSM (IPM) 110 kW, 400 V, 180 A, $f_{1,max} = 200$ Hz, $L_q \approx 0.3$ pu, $f_{sw} = 8$ kHz |
|---|---|---|---|
| M1 | cable (5.1): $v = 150$ m/µs, $t_r = 0.2$ µs, $l = 150$ m, $\Gamma_m = 0.9$ | $l_{crit} = 15$ m $\Rightarrow \hat{V}_m \approx 1.9 V_{dc} = 1235$ V with $V_{dc} = 650$ V: at the MG 1-31 limit (1240 V), above MG 1-30 (1000 V) $\Rightarrow$ filter needed; a $dv/dt$ filter for $OF \le 0.2$ would need $t_r \ge 9$ µs ($f_0 \le 33$ kHz) with $C_f \le 0.22$ µF (loss bound 1.1 kW at 4 kHz); the sine filter is chosen | |
| M2 | window (5.2) | $8 f_{1,max} = 600$ Hz $\le f_0 \le f_{sw}/4 = 1$ kHz | 1.6 kHz $\le f_0 \le$ 2 kHz (infeasible at 4 kHz) |
| M3 | $L_f$ from (5.3) | $\epsilon_L \le 0.08$ at 75 Hz $\Rightarrow L_f \le 201$ µH; $L_f = 230$ µH (6.1 %): drop 6.1 % at 50 Hz, 9.2 % at 75 Hz; $\Delta i_{\mathrm{pp},max} = 118$ A (43 %), RMS 17.8 A | $L_f \le 82$ µH would give 65 % ripple $\Rightarrow L_f = 150$ µH (3.7 %): drop 14.7 % at 200 Hz (budgeted in $V_{dc}$); $\Delta i_{\mathrm{pp},max} = 90$ A (35 %), RMS 13.6 A |
| M4 | $C_f$ from (5.4) | $f_0 \to 900$ Hz: $C_f = 140$ µF (5.2 % $C_b$), $f_0 = 887$ Hz; $I_c = 10.2/15.2$ A at 50/75 Hz (8 %) | $f_0 \to 1.8$ kHz: $C_f = 50$ µF (2 % $C_b$), $f_0 = 1838$ Hz; $I_c = 14.5$ A at 200 Hz (8 %) |
| M5 | loaded resonance (3.4) | $L_{hf} = 0.68$ mH: $f_p = 1.16 f_0 = 1026$ Hz (at the window edge), $\lvert v_c/v_i \rvert_{f_{sw}} = 0.070$ (−23 dB), $\zeta_m \approx 0.003$ | $L_{hf} = 1.23$ mH: $f_p = 1.06 f_0 = 1947$ Hz, $\lvert v_c/v_i \rvert_{f_{sw}} = 0.063$ (−24 dB), $\zeta_m \approx 10^{-4}$ |
| M6 | damping | series $R_d = 0.37\,\Omega$ ($\zeta \approx 0.17$): 607 W (0.55 %) at 75 Hz; R–C $n = 1$: +140 µF, $R_2 = 1.66\,\Omega$, 1.19 kW (1.1 %) | series $R_d = 0.55\,\Omega$: 648 W (0.6 %) at 200 Hz; R–C $n = 1$: +50 µF, $R_2 = 2.45\,\Omega$, 1.55 kW (1.4 %) $\Rightarrow$ active damping preferred |
| M7 | control, Section 5.6 | $\alpha_i \le 2\pi \cdot 400$ Hz, stator-current loop $\le f_p/5 \approx 200$ Hz, $f_{inj} \le 340$ Hz | $\alpha_i \le 2\pi \cdot 800$ Hz, stator-current loop $\le 390$ Hz, $f_{inj} \le 650$ Hz; $L_d \ne L_q$ in the observer |

The two columns show the trade-off: the induction motor at 4 kHz fits the window with a 6 % inductor and a 5 % capacitor, the PMSM needs 8 kHz and still pays a 15 % drop at its top speed or a larger ripple; in both cases the passive damping costs 0.5–1.5 % of the rated power, which is why the resonance is damped actively whenever the controller has the filter in its model. The dead-time harmonics must be checked against $f_p$ at every operating speed (the 7th of 75 Hz is 525 Hz, that of 200 Hz is 1.4 kHz, both below $f_p$ with margin), and the saturation of $L_q$ at rated $i_q$ shifts $f_p$ of the IPM upwards by a few percent more.

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